

Mathematical Guide to the KGPardis src/Eshelby* Implementations

Source-level technical documentation

Code snapshot reviewed 7 August 2026

Abstract

This document explains every source file matching `src/Eshelby*` in the reviewed KGPardis tree. It gives the equations represented by the packed arrays, relates each loop and matrix operation to a Cartesian multi-index formula, and derives the direct segment–inclusion nodal force. Equations labelled “literal implementation” describe what the present source computes; where that differs from the recentering identity or from another code path, the difference is stated explicitly rather than silently corrected.

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1 Scope and computational architecture

There are eight matching files. Seven implement the Eshelby-specific fast multipole method (FMM), while one evaluates a nearby inclusion's force on a straight dislocation segment by analytic line integration.

File	Role	Guard
<code>src/EshelbyBuildMatrix.cc</code>	Reconstructs a rectangular contraction block from a packed fully symmetric tensor.	none
<code>src/EshelbyDerofR.cc</code>	Generates all unique Cartesian derivatives of the inverse-distance kernel needed by M2L.	none
<code>src/EshelbyEvalTaylor.cc</code>	Evaluates a local Taylor expansion at a point and returns a symmetric stress matrix.	ESHELBY
<code>src/EshelbyForce.cc</code>	Integrates the inclusion-induced line-force density against the two linear nodal shape functions.	both
<code>src/EshelbyMakeEta.cc</code>	Forms raw monomial multipole moments for one spherical inclusion; it is also reused to form displacement monomials.	none
<code>src/EshelbyMakeTaylor.cc</code>	Performs the multipole-to-local (M2L) operation.	ESHELBY
<code>src/EshelbyTranslateEta.cc</code>	Performs a hard-coded multipole-to-multipole (M2M) binomial translation through order six.	ESHELBY
<code>src/EshelbyTranslateTaylor.cc</code>	Performs the local-to-local (L2L) translation used in the downward FMM pass.	ESHELBY

Here “both” means that `src/EshelbyForce.cc` is nested inside `ESHELBY` and `ESHELBYFORCE`. The far-field sequence is

$$\boxed{\text{inclusion} \xrightarrow{\text{MakeEta}} M \xrightarrow{\text{TranslateEta}} M' \xrightarrow[\text{BuildMatrix}]{\text{DerofR} + \text{MakeTaylor}} L \xrightarrow{\text{TranslateTaylor}} L' \xrightarrow{\text{EvalTaylor}} \sigma(\mathbf{x}).}$$

The direct path `src/EshelbyForce.cc` bypasses these expansions and returns the two endpoint forces directly.

The global setup limits the sum of multipole and Taylor orders to

$$p + t \leq \text{MAX_ORDER_ESHELBY} = 6$$

in `include/FM.h:14` and `src/FMComm.cc:1464–1485`. This is essential because the derivative and translation tables stop at that order.

2 Shared mathematical representation

2.1 Multi-indices and packed symmetric tensors

For a Cartesian multi-index

$$\alpha = (\alpha_x, \alpha_y, \alpha_z) \in \mathbb{N}_0^3, \quad |\alpha| = \alpha_x + \alpha_y + \alpha_z,$$

write

$$\mathbf{x}^\alpha = x^{\alpha_x} y^{\alpha_y} z^{\alpha_z}, \quad \alpha! = \alpha_x! \alpha_y! \alpha_z!, \quad \binom{\alpha}{\beta} = \prod_{q \in \{x, y, z\}} \binom{\alpha_q}{\beta_q}.$$

The shared table at `src/FMComm.cc:78` is

$$\text{sequence}[n] = S_n = \binom{n+2}{3}.$$

It is the number of unique components in all symmetric ranks below (n). Consequently, a symmetric rank-(n) tensor has

$$N_n = S_{n+1} - S_n = \binom{n+2}{2}$$

unique entries and occupies the half-open global interval $[S_n, S_{n+1})$.

For $\alpha = (a, b, c)$, $a + b + c = n$, define $u = b + c$. Its local and global packed indices are

$$\ell_n(a, b, c) = \frac{u(u+1)}{2} + c, \quad I(\alpha) = S_n + \ell_n(a, b, c). \quad (1)$$

Thus the order is

$$\begin{array}{ll} n & \text{packed components} \\ 0 & 1 \\ 1 & x, y, z \\ 2 & x^2, xy, xz, y^2, yz, z^2 \\ 3 & x^3, x^2y, x^2z, xy^2, xyz, xz^2, y^3, y^2z, yz^2, z^3. \end{array}$$

The comments made of digits in the source use $1 = x$, $2 = y$, and $3 = z$. For example, `1 1 2 3` means the x^2yz component.

Moment vectors through degree p contain S_{p+1} entries. Local Taylor vectors omit ranks zero and one and store ranks two through $t + 2$; their local packed index and total length are

$$J(\alpha) = I(\alpha) - 4, \quad 2 \leq |\alpha| \leq t + 2, \quad N_L = S_{t+3} - 4. \quad (2)$$

In particular, the first six Taylor entries are ordered

$$(xx, xy, xz, yy, yz, zz).$$

2.2 Factorial weights

The raw monomial and moment arrays do not include factorial denominators. The array `multFactor`, initialized in `src/FMComm.cc:149–255`, contains

$$\text{multFactor}[I(\alpha) - S_2] = \frac{1}{\alpha!}, \quad 2 \leq |\alpha| \leq 6. \quad (3)$$

Degrees zero and one need no stored factor because their factorials are one. The M2L, L2L, and evaluation routines apply these factors immediately before a matrix–vector product.

2.3 The far-field kernel

Let $r = \|\mathbf{r}\|$. The FMM is built from the symmetric tensor

$$K_{ij}(\mathbf{r}) = \frac{\delta_{ij}}{r^3} - \frac{3r_i r_j}{r^5} = -\partial_i \partial_j \frac{1}{r}. \quad (4)$$

For a spherical inclusion of radius (a) and volumetric strain

$$E_v = \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz},$$

the scalar source strength and isotropic material factor used by the FMM are

$$q = E_v a^3, \quad A(\mu, \nu) = \frac{\mu(1 + \nu)}{1 - 2\nu}. \quad (5)$$

The monopole field represented by that path is therefore

$$\sigma_{ij}^{\text{FMM}}(\mathbf{r}) = AqK_{ij}(\mathbf{r}) = AE_v a^3 \left(\frac{\delta_{ij}}{r^3} - \frac{3r_i r_j}{r^5} \right). \quad (6)$$

No factor $4\pi/3$ is stored: $E_v a^3$, not geometric volume times strain, is the source normalization in this implementation.

3 EshelbyMakeEta.cc: particle-to-multipole moments

The implementation is at `src/EshelbyMakeEta.cc:35–166`. With inclusion position $\mathbf{x}_s$, expansion center $\mathbf{c}$, radius $a = R_0$, and requested order p , it first forms

$$\mathbf{d} = \mathbf{c} - \mathbf{x}_s, \quad q = E_v R_0^3. \quad (7)$$

The assignments at lines 56–159 enumerate every raw monomial through degree six. The final scaling loop at lines 161–163 makes the returned coefficient for every $|\alpha| \leq p$

$$\boxed{M_\alpha = q \mathbf{d}^\alpha}. \quad (8)$$

For example,

$$\begin{aligned} (M_0, M_x, M_y, M_z) &= q(1, d_x, d_y, d_z), \\ (M_{xx}, M_{xy}, M_{xz}, M_{yy}, M_{yz}, M_{zz}) &= q(d_x^2, d_x d_y, d_x d_z, d_y^2, d_y d_z, d_z^2). \end{aligned}$$

There is deliberately no $1/\alpha!$ in (8); callers apply (3) when a Taylor theorem requires it.

For many inclusions in one leaf cell, `src/CellCharge.cc:613–650` sums these coefficients, so the cell moment is

$$M_\alpha = \sum_s E_{v,s} a_s^3 (\mathbf{c} - \mathbf{x}_s)^\alpha. \quad (9)$$

That call site also rejects unequal radii for the FMM. The file’s contract is $0 \leq p \leq 6$; the function itself does not validate the order.

4 EshelbyDerofR.cc: derivatives of the kernel

The routine at `src/EshelbyDerofR.cc:22–856` accepts a nonzero separation vector $\mathbf{x}$ and fills all distinct components of

$$D_{i_1 \dots i_n}(\mathbf{x}) = -\partial_{i_1} \cdots \partial_{i_n} \frac{1}{r}, \quad 2 \leq n \leq \text{order} + 2. \quad (10)$$

Rank two is exactly K_{ij} from (4). A compact formula covering all hard-coded entries is

$$\begin{aligned} D_{i_1 \dots i_n} &= \sum_{p=0}^{\lfloor n/2 \rfloor} (-1)^{n-p+1} (2n - 2p - 1)!! r^{-(2n-2p+1)} \\ &\quad \times \sum_{\mathcal{P}_p} \left(\prod_{\{a,b\} \in \mathcal{P}_p} \delta_{a i_b} \right) \left(\prod_{c \notin V(\mathcal{P}_p)} x_{i_c} \right), \end{aligned} \quad (11)$$

where $\mathcal{P}_p$ ranges over all choices of p disjoint pairs of index positions, and $V(\mathcal{P}_p)$ is the set of paired positions.

The first two nontrivial ranks illustrate the coefficients encoded by the source:

$$D_{ijk} = -\frac{3}{r^5}(\delta_{ij}x_k + \delta_{ik}x_j + \delta_{jk}x_i) + \frac{15x_ix_jx_k}{r^7}, \quad (12)$$

$$D_{ijkl} = -\frac{3}{r^5}(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + \frac{15}{r^7}\text{Sym}_6(\delta_{ij}x_kx_l) - \frac{105x_ix_jx_kx_l}{r^9}. \quad (13)$$

Here Sym_6 denotes the six distinct placements of one Kronecker delta and two coordinates.

The inverse-radius cache obeys

$$\text{dinv}[0] = r^{-1}, \quad \text{dinv}[m] = (2m-1)!!r^{-(2m+1)}, \quad 1 \leq m \leq 8, \quad (14)$$

and `xv` stores the monomials at index $I(\alpha)$. Since `drList` omits ranks zero and one,

$$\boxed{\text{drList}[S_n - 4 + \ell_n(\alpha)] = D_\alpha, \quad n = |\alpha| \geq 2.} \quad (15)$$

Condition	New rank	<code>drList</code> indices	Source lines
always	2	0–5	57–62
<code>order > 0</code>	3	6–15	64–89
<code>order > 1</code>	4	16–30	92–162
<code>order > 2</code>	5	31–51	164–259
<code>order > 3</code>	6	52–79	262–399
<code>order > 4</code>	7	80–115	402–592
<code>order > 5</code>	8	116–160	594–851

The required length is

$$S_{\text{order}+3} - 4 = \binom{\text{order} + 5}{3} - 4.$$

An independent symbolic check of all 161 hard-coded components through rank eight found agreement with (10) to floating-point roundoff at a generic nonsymmetric point. There is no $r = 0$ guard; separated FMM cells are an input precondition.

5 EshelbyBuildMatrix.cc: unpacking contraction blocks

This utility is implemented at `src/EshelbyBuildMatrix.cc:40–87`. For arguments called Taylor order t and multipole order m , its active matrix has

$$n_r = N_{t+2} = \binom{t+4}{2}, \quad n_c = N_m = \binom{m+2}{2}. \quad (16)$$

Rows correspond to $|\alpha| = t + 2$, columns to $|\beta| = m$, and the mathematical operation is simply

$$\boxed{B_{\alpha\beta} = D_{\alpha+\beta}.} \quad (17)$$

More explicitly, the row loops at lines 69–83 enumerate

$$\alpha = (t+2-i, i-j, j), \quad r_{\text{row}} = \frac{i(i+1)}{2} + j, \quad 0 \leq j \leq i \leq t+2,$$

and the column loops enumerate

$$\beta = (m - k, k - l, l), \quad c_{\text{col}} = \frac{k(k+1)}{2} + l, \quad 0 \leq l \leq k \leq m.$$

The source's running offset simplifies to

$$\text{drOffset} = \frac{(i+k)(i+k+1)}{2} + j + l = \ell_{t+m+2}(\alpha + \beta),$$

which proves (17). Storage is row-major with caller-provided leading dimension `mCols`:

$$\text{matrix}[r \text{ mCols} + c].$$

Only the active upper-left rectangle is cleared; padding in an oversized work array is left unchanged. Although the argument is named `drList`, the routine is generic for this packed layout: the Taylor evaluation and L2L paths also pass a packed Taylor tensor to it. It applies neither a factorial nor a material factor.

6 EshelbyMakeTaylor.cc: multipole-to-local conversion

The M2L routine is at `src/EshelbyMakeTaylor.cc:30–124`. Let $\mathbf{c}_s$ be the source-cell center, $\mathbf{c}_t$ the target-cell center, and

$$\mathbf{R} = \mathbf{c}_t - \mathbf{c}_s.$$

This direction is established by the production call at `src/CellCharge.cc:284–326`. For every local derivative multi-index α , $|\alpha| \leq t$, the generated Taylor tensor is

$$L_{ij,\alpha} = A(\mu, \nu) \sum_{|\beta| \leq p} D_{ij,\alpha+\beta}(\mathbf{R}) \frac{M_\beta}{\beta!}. \quad (18)$$

All free indices are symmetric, so $L_{ij,\alpha}$ is stored as one packed tensor component of rank $|\alpha| + 2$, using (2).

The source realizes (18) as follows.

1. Lines 43–48 allocate $S_{p+t+3} - 4$ entries and call `EshelbyDerofR` through derivative rank $p + t + 2$.
2. Lines 50–86 assemble every $(|\alpha| + 2, |\beta|)$ block with `EshelbyBuildMatrix`. The complete rectangular matrix has

$$(S_{t+3} - 4) \text{ rows and } S_{p+1} \text{ columns.}$$

3. Lines 91–105 copy M_β and multiply degrees two and above by $1/\beta!$. The `multFactor` table must already have been initialized by `EshelbyInitFMArray`.
4. Lines 109–114 perform the matrix–vector product and multiply every output by

$$A(\mu, \nu) = \frac{\mu(1 + \nu)}{1 - 2\nu}.$$

The separation must be nonzero and $p + t \leq 6$. The material factor is singular at $\nu = 1/2$.

7 EshelbyTranslateEta.cc: multipole translation

7.1 The binomial contraction represented by the table

The M2M implementation is at `src/EshelbyTranslateEta.cc:27–649`. It first calls

$$\text{EshelbyMakeEta}(1, 1, \text{newCenter}, \text{oldCenter}, p, \dots),$$

so, by (7), its literal displacement is

$$\boldsymbol{\delta} = \mathbf{c}_{\text{old}} - \mathbf{c}_{\text{new}}. \quad (19)$$

The outer product at lines 42–54 constructs

$$\mathbf{mat}_{\beta\gamma} = M_{\beta} \boldsymbol{\delta}^{\gamma}.$$

The long rank-by-rank table then implements the symmetric-tensor binomial convolution

$$\boxed{\widetilde{M}_{\alpha} = \sum_{0 \leq \beta \leq \alpha} \binom{\alpha}{\beta} M_{\beta} \boldsymbol{\delta}^{\alpha-\beta}, \quad |\alpha| \leq p,} \quad (20)$$

apart from the single table entry discussed below. The implemented blocks are rank zero and one at lines 56–64, rank two at 67–79, rank three at 82–125, rank four at 130–217, rank five at 220–373, and rank six at 377–646.

For example, the rank-two and pure rank-three formulas are

$$\begin{aligned} \widetilde{M}_{xx} &= M_{xx} + 2M_x \delta_x + M_0 \delta_x^2, \\ \widetilde{M}_{xxx} &= M_{xxx} + 3M_{xx} \delta_x + 3M_x \delta_x^2 + M_0 \delta_x^3. \end{aligned}$$

The explicit coefficients 2, 3, 6, 10, 15, 20, ... are the multi-index binomial coefficients in (20).

7.2 Literal direction versus the recentering identity

Moments constructed by `EshelbyMakeEta` satisfy

$$M_{\alpha}^{\text{old}} = \sum_s q_s (\mathbf{c}_{\text{old}} - \mathbf{x}_s)^{\alpha}.$$

Direct construction at the new center therefore requires

$$M_{\alpha}^{\text{new}} = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} M_{\beta}^{\text{old}} \mathbf{h}^{\alpha-\beta}, \quad \mathbf{h} = \mathbf{c}_{\text{new}} - \mathbf{c}_{\text{old}}. \quad (21)$$

The implementation uses $\boldsymbol{\delta} = -\mathbf{h}$, reversing all terms with odd shift degree. This is not just a naming ambiguity: the upward-pass call at `src/FMComm.cc:2282–2294` passes the parent as the new center and the child as the old center, consistent with the function comment. At first order, the distinction is already visible:

$$\underbrace{M_x^{\text{old}} + (c_{\text{new},x} - c_{\text{old},x})M_0}_{\text{direct recentering}} \quad \text{versus} \quad \underbrace{M_x^{\text{old}} + (c_{\text{old},x} - c_{\text{new},x})M_0}_{\text{literal source}}.$$

7.3 Table audit and resource behavior

An equation-by-equation audit of all 80 explicit rank-two through rank-six updates found one index mismatch and no other coefficient/index mismatch. For `ep[41]`, the x^2y^3 component, line 268 contains

$$\text{mat}(xx, yyy) + \text{mat}(yyy, xz),$$

where the symmetric second term required by (20) is

$$\text{mat}(xx, yyy) + \text{mat}(yyy, xx).$$

In source indices, `ELEM(16,6,nCols)` would have to be `ELEM(16,4,nCols)`. As written, this one term combines to the wrong multi-index.

The arrays `etaTrans` and `mat` allocated at lines 37 and 47 are not freed before the return at line 648, so every call leaks both buffers. Input and output may otherwise alias safely because the full outer product is built before `ep` is written.

8 EshelbyTranslateTaylor.cc: local translation

The L2L routine is at `src/EshelbyTranslateTaylor.cc:25–126`. For output derivative order $i = 0, \dots, t$ and displacement degree $j = 0, \dots, t - i$, it asks `EshelbyBuildMatrix` for

$$B_{\alpha\beta}^{(i,j)} = L_{\alpha+\beta}, \quad |\alpha| = i + 2, \quad |\beta| = j. \quad (22)$$

The row and column block offsets at lines 72–84 are $S_{i+2} - 4$ and S_j , respectively. Lines 105–117 then multiply each disjoint row block by all admissible displacement monomials. With correctly normalized monomials, the literal displacement from (19) gives

$$\tilde{L}_\alpha = \sum_{|\beta| \leq t-i} L_{\alpha+\beta} \frac{\delta^\beta}{\beta!}, \quad |\alpha| = i + 2. \quad (23)$$

On the other hand, `EshelbyEvalTaylor` uses powers of $\mathbf{x} - \mathbf{c}$. The mathematical recentering identity for a polynomial from $\mathbf{c}_{\text{old}}$ to $\mathbf{c}_{\text{new}}$ is therefore

$$L_\alpha^{\text{new}} = \sum_{|\beta| \leq t-i} L_{\alpha+\beta}^{\text{old}} \frac{\mathbf{h}^\beta}{\beta!}, \quad \mathbf{h} = \mathbf{c}_{\text{new}} - \mathbf{c}_{\text{old}}. \quad (24)$$

The file again uses $\delta = -\mathbf{h}$. The downward call at `src/FMComm.cc:2074–2125` passes the parent center as old and child center as new, so odd shift-degree terms have the opposite sign from (24) under the documented evaluation convention.

There is also a precise endpoint error in factorial scaling. Lines 51–55 set the last valid index to $S_{t+1} - 1$ but loop while the index is strictly less than that value. For every $t \geq 2$, all needed monomials are divided by their factorial except the final packed component:

$$q_\beta = \frac{\delta^\beta}{\beta!} \quad \text{except} \quad q_{(0,0,t)} = \delta_z^t. \quad (25)$$

Thus only the output rank-two block can receive the highest-degree error, with excess

$$\left(1 - \frac{1}{t!}\right) L_{\alpha+(0,0,t)} \delta_z^t, \quad |\alpha| = 2.$$

The temporary matrix is freed at lines 89–90, and the two remaining buffers are freed at lines 119–123.

9 EshelbyEvalTaylor.cc: local expansion at a point

The implementation is at `src/EshelbyEvalTaylor.cc:24–100`. Let

$$\mathbf{y} = \mathbf{x} - \mathbf{c}_t.$$

The call at line 41 deliberately reverses the apparently descriptive arguments to `EshelbyMakeEta`:

$$\text{position} = \mathbf{c}_t, \quad \text{center} = \mathbf{x},$$

which correctly produces raw powers $\mathbf{y}^\alpha$. Lines 69–75 apply every $1/\alpha!$ factor, including the final component. The evaluated stress is

$$\sigma_{ij}(\mathbf{x}) = \sum_{|\alpha| \leq t} L_{ij,\alpha} \frac{\mathbf{y}^\alpha}{\alpha!}. \quad (26)$$

For each degree, `EshelbyBuildMatrix(taylorSeries,0,i,...)` reconstructs the six stress rows $ij \in (xx, xy, xz, yy, yz, zz)$ from the packed fully symmetric local tensor. After the matrix–vector product, lines 82–90 return

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz} \end{pmatrix}. \quad (27)$$

The allocated matrix and output vector have $S_{t+3} - 4$ rows even though only their first six rows are populated or consumed; this is extra work and storage, not a change to (26).

10 EshelbyForce.cc: analytic endpoint force

10.1 Inputs and model restrictions

The helpers are at `src/EshelbyForce.cc:31–103` and the public routine at lines 134–287. It uses the inclusion center $\mathbf{c}$, three radius entries, the six-component symmetric strain vector, two segment endpoints, and the Burgers vector $\mathbf{b}$. The strain layout is

$$(\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{zz}, \varepsilon_{yz}, \varepsilon_{xz}, \varepsilon_{xy}),$$

but the routine uses only

$$E_v = \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz}.$$

The source comment assumes the three diagonal entries are equal, although the code checks neither equality nor absence of deviatoric diagonal strain. All shear entries are ignored.

Although the inclusion type stores three semi-principal axes, this routine sets

$$a = \max(a_x, a_y, a_z) \quad (28)$$

and performs a spherical intersection. It does not receive or use the ellipsoid orientation. Matrix properties are (μ_1, ν_1) , while inclusion properties are (μ_2, ν_2) ; initialization makes material 2 equal to material 1 if no second values were supplied.

10.2 Segment geometry and sphere intersection

Let the endpoints be $\mathbf{x}_1, \mathbf{x}_2$, and define

$$\boldsymbol{\ell} = \mathbf{x}_2 - \mathbf{x}_1, \quad L = \|\boldsymbol{\ell}\|, \quad \mathbf{t} = \frac{\boldsymbol{\ell}}{L}. \quad (29)$$

With $\mathbf{R}_i = \mathbf{x}_i - \mathbf{c}$, the axial coordinates and closest approach vector are

$$s_0 = \mathbf{R}_1 \cdot \mathbf{t}, \quad s_1 = \mathbf{R}_2 \cdot \mathbf{t} = s_0 + L, \quad \mathbf{n} = \mathbf{R}_1 - s_0 \mathbf{t}, \quad d^2 = \|\mathbf{n}\|^2. \quad (30)$$

Hence a segment point and its distance from the inclusion center are

$$\mathbf{r}(s) = \mathbf{n} + s\mathbf{t}, \quad \rho(s) = \sqrt{d^2 + s^2}, \quad s \in [s_0, s_1]. \quad (31)$$

The infinite line intersects the effective sphere when $\Delta = a^2 - d^2 > 0$, at

$$s_- = -\sqrt{\Delta}, \quad s_+ = +\sqrt{\Delta}. \quad (32)$$

Lines 209–244 divide the segment into exterior set $\mathcal{O}$ and interior set $\mathcal{I}$:

Case	$\mathcal{O}$	$\mathcal{I}$
line miss or no segment overlap	$[s_0, s_1]$	$\emptyset$
starts inside, exits	$[s_+, s_1]$	$[s_0, s_+]$
enters, ends inside	$[s_0, s_-]$	$[s_-, s_1]$
wholly inside	$\emptyset$	$[s_0, s_1]$
passes through	$[s_0, s_-] \cup [s_+, s_1]$	$[s_-, s_+]$

For two exterior pieces, their five integral vectors are added componentwise.

10.3 Interior and exterior primitive integrals

For an interior interval $[p, q]$, `EshelbySpecialIntegrals` returns

$$S_0 = \int_p^q 1 \, ds = q - p, \quad S_1 = \int_p^q s \, ds = \frac{q^2 - p^2}{2}. \quad (33)$$

For an exterior interval, `EshelbyIntegrals` returns endpoint differences of the following antiderivatives:

$$\begin{aligned} \Phi_{03}(s) &= \frac{s}{d^2 \rho}, & \Phi_{13}(s) &= -\frac{1}{\rho}, \\ \Phi_{05}(s) &= \frac{2\Phi_{03}(s) + s/\rho^3}{d^2}, & \Phi_{15}(s) &= -\frac{1}{\rho^3}, \\ \Phi_{25}(s) &= \Phi_{03}(s) - \frac{s}{\rho^3}. \end{aligned} \quad (34)$$

Thus the array `wInt` has the exact meanings

$$\begin{aligned} W_0 &= \int_{\mathcal{O}} \rho^{-3} \, ds, & W_1 &= \int_{\mathcal{O}} s \rho^{-3} \, ds, \\ W_2 &= 3 \int_{\mathcal{O}} \rho^{-5} \, ds, & W_3 &= 3 \int_{\mathcal{O}} s \rho^{-5} \, ds, \\ W_4 &= 3 \int_{\mathcal{O}} s^2 \rho^{-5} \, ds. \end{aligned} \quad (35)$$

The factors of three in the last three entries are part of the array definition.

If

$$d^2 < 10^{-8}(p^2 + q^2),$$

the source uses the collinear limit

$$\Phi_{03} = -\frac{s}{2\rho^3}, \quad \Phi_{13} = -\rho^{-1}, \quad \Phi_{05} = \Phi_{15} = \Phi_{25} = 0.$$

At exactly $d = 0$, the zeroed terms multiply $\mathbf{n} \times \mathbf{t} = \mathbf{0}$; for small nonzero d , this is an approximation. The code computes $1/d^2$ before taking this branch, so an exactly collinear case still produces an unused floating-point infinity.

10.4 Shape functions and endpoint moments

The linear segment shape functions are

$$N_1(s) = \frac{s_1 - s}{L}, \quad N_2(s) = \frac{s - s_0}{L}. \quad (36)$$

For endpoint 2, lines 258–261 form

$$\begin{aligned} F_{03}^{(2)} &= W_1 - s_0 W_0, & F_{05}^{(2)} &= W_3 - s_0 W_2, \\ F_{15}^{(2)} &= W_4 - s_0 W_3, & F_{\text{SP}}^{(2)} &= S_1 - s_0 S_0. \end{aligned} \quad (37)$$

For endpoint 1, lines 275–278 form

$$\begin{aligned} F_{03}^{(1)} &= s_1 W_0 - W_1, & F_{05}^{(1)} &= s_1 W_2 - W_3, \\ F_{15}^{(1)} &= s_1 W_3 - W_4, & F_{\text{SP}}^{(1)} &= s_1 S_0 - S_1. \end{aligned} \quad (38)$$

For example,

$$F_{05}^{(2)} = 3 \int_{\mathcal{O}} (s - s_0) \rho^{-5} ds, \quad F_{\text{SP}}^{(1)} = \int_{\mathcal{I}} (s_1 - s) ds.$$

Division by L is applied later through the material factors.

10.5 Vector and material contraction

The three fixed vectors at lines 246–256 are

$$\mathbf{A} = \mathbf{b} \times \mathbf{t}, \quad \mathbf{B} = (\mathbf{b} \cdot \mathbf{n})(\mathbf{n} \times \mathbf{t}), \quad \mathbf{D} = (\mathbf{b} \cdot \mathbf{t})(\mathbf{n} \times \mathbf{t}). \quad (39)$$

Define

$$C_j = \frac{\mu_j(1 + \nu_j)}{1 - 2\nu_j} E_v, \quad j \in \{1, 2\}. \quad (40)$$

The exact endpoint equations implemented at lines 270–283 are

$$\boxed{\mathbf{f}_i = -\frac{C_1 a^3}{L} \left(\mathbf{A} F_{03}^{(i)} + \mathbf{B} F_{05}^{(i)} + \mathbf{D} F_{15}^{(i)} \right) + \frac{2C_2}{L} \mathbf{A} F_{\text{SP}}^{(i)}, \quad i = 1, 2.} \quad (41)$$

Equivalently, this is the shape-function integral of the literal line-force density

$$\mathbf{g}_{\text{out}}(s) = -C_1 a^3 \left[\frac{\mathbf{b} \times \mathbf{t}}{\rho^3} + 3 \frac{(\mathbf{b} \cdot \mathbf{r})(\mathbf{r} \times \mathbf{t})}{\rho^5} \right], \quad (42)$$

$$\mathbf{g}_{\text{in}}(s) = 2C_2(\mathbf{b} \times \mathbf{t}), \quad (43)$$

$$\mathbf{f}_i = \int_{\mathcal{O}} N_i \mathbf{g}_{\text{out}} \, ds + \int_{\mathcal{I}} N_i \mathbf{g}_{\text{in}} \, ds. \quad (44)$$

Both nodal forces are perpendicular to $\mathbf{t}$.

Using the standard Peach–Koehler relation $\mathbf{g} = (\boldsymbol{\sigma} \mathbf{b}) \times \mathbf{t}$, the stress fields algebraically implied by this file are

$$\boxed{\boldsymbol{\sigma}_{\text{out}}^{\text{force file}} = -C_1 a^3 \left(\frac{\mathbf{I}}{\rho^3} + 3 \frac{\mathbf{r} \mathbf{r}^T}{\rho^5} \right), \quad \boldsymbol{\sigma}_{\text{in}}^{\text{force file}} = 2C_2 \mathbf{I}.} \quad (45)$$

11 Cross-file consistency and implementation cautions

This section records literal source behavior that is easy to miss when reading only the comments.

11.1 Direct-force and FMM fields are not the same kernel

The FMM monopole (6) is

$$\boldsymbol{\sigma}_{\text{out}}^{\text{FMM}} = +C_1 a^3 \left(\frac{\mathbf{I}}{\rho^3} - 3 \frac{\mathbf{r} \mathbf{r}^T}{\rho^5} \right).$$

The direct routine implies (45), which differs in the overall isotropic sign and in the relative sign of the radial term. Its interior value also has the opposite sign from the disabled reference stress at `src/RemoteSegForces.cc:290–335`, which states $-2C_1 \mathbf{I}$. The material-2 constants can intentionally alter the interior magnitude, but do not explain these sign differences. Therefore near-field direct and far-field FMM inclusion forces are not algebraically consistent under the Peach–Koehler convention used by `src/RemoteSegForces.cc:903–905`. A numerical regression test spanning the near/far cutoff is warranted before choosing which formula to change.

11.2 Translation discrepancies

Both translation functions obtain $\mathbf{c}_{\text{old}} - \mathbf{c}_{\text{new}}$ from `EshelbyMakeEta`, while the moment definition and Taylor evaluation require $\mathbf{c}_{\text{new}} - \mathbf{c}_{\text{old}}$. This reverses all odd shift-degree contributions in M2M and L2L. In addition:

- `src/EshelbyTranslateEta.cc:268` has one wrong packed matrix index in the $x^2 y^3$ coefficient;
- that function never frees its two temporary heap arrays;
- `src/EshelbyTranslateTaylor.cc:51–55` fails to divide the final z^t displacement monomial by $t!$.

11.3 Input and boundary assumptions

- The FMM tables require nonnegative orders with $p + t \leq 6$, but the individual functions generally rely on the global setup rather than checking these bounds locally.
- `EshelbyDerofR` divides by r , and `EshelbyForce` divides by segment length L ; neither accepts a zero value.
- All factors containing $1 - 2\nu$ assume $\nu \neq 1/2$.
- The direct-force routine replaces any ellipsoid by a sphere of maximum semi-axis, whereas the FMM rejects unequal radii.

- Exact, strict intersection comparisons at `src/EshelbyForce.cc:209–235` make an endpoint exactly on the sphere a delicate boundary case. Certain “boundary to interior” configurations fall into the final through-going branch and can create a reversed exterior interval.
- The no-overlap condition uses bitwise `|` rather than logical `||`. Its truth value is the same for ordinary comparison results, but it evaluates all three operands.

12 Compact end-to-end equation summary

For a well-separated source cell and target cell, the nominal mathematical pipeline represented by the files can be summarized as

$$M_{\beta}^{(s)} = \sum_a E_{v,a} a_a^3 (\mathbf{c}_s - \mathbf{x}_a)^{\beta}, \quad \text{P2M: EshelbyMakeEta,} \quad (46)$$

$$M_{\alpha}^{(p)} = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} M_{\beta}^{(s)} (\mathbf{c}_p - \mathbf{c}_s)^{\alpha - \beta}, \quad \text{M2M: intended recentering,} \quad (47)$$

$$L_{ij,\alpha} = A \sum_{|\beta| \leq p} D_{ij,\alpha+\beta} (\mathbf{c}_t - \mathbf{c}_s) \frac{M_{\beta}}{\beta!}, \quad \text{M2L: DerofR/BuildMatrix/MakeTaylor,} \quad (48)$$

$$L_{ij,\alpha}^{(c)} = \sum_{|\beta| \leq t - |\alpha|} L_{ij,\alpha+\beta}^{(p)} \frac{(\mathbf{c}_c - \mathbf{c}_p)^{\beta}}{\beta!}, \quad \text{L2L: intended recentering,} \quad (49)$$

$$\sigma_{ij}(\mathbf{x}) = \sum_{|\alpha| \leq t} L_{ij,\alpha} \frac{(\mathbf{x} - \mathbf{c}_t)^{\alpha}}{\alpha!}, \quad \text{L2P: EshelbyEvalTaylor.} \quad (50)$$

Equations (47) and (49) are the nominal recenterings. The present translation sources use the negatives of the shown center displacements, as detailed in their respective sections.

A Source-to-equation index

File and lines	Main equations	Principal arrays/outputs
<code>src/EshelbyBuildMatrix.cc:40–87</code>	(16), (17)	Rectangular row-major block $B_{\alpha\beta}$.
<code>src/EshelbyDerofR.cc:22–856</code>	(10)–(15)	Packed ranks 2–8 in <code>drList</code> ; monomial cache <code>xv</code> .
<code>src/EshelbyEvalTaylor.cc:24–100</code>	(26), (27)	Symmetric 3×3 <code>stressMatrix</code> .
<code>src/EshelbyForce.cc:31–103, 134–287</code>	(31), (35), (41)	Five exterior integrals, two interior integrals, endpoint forces <code>f1/f2</code> .
<code>src/EshelbyMakeEta.cc:35–166</code>	(7)–(9)	Raw moment/monomial vector <code>etav</code> .
<code>src/EshelbyMakeTaylor.cc:30–124</code>	(18)	Packed local tensor <code>taylorSeries</code> .
<code>src/EshelbyTranslateEta.cc:27–649</code>	(20), (21)	Translated moments <code>ep</code> ; explicit ranks 0–6.
<code>src/EshelbyTranslateTaylor.cc:25–126</code>	(22)–(25)	Translated local tensor <code>tPrime</code> .

B Notation quick reference

Symbol	Meaning
$S_n = \binom{n+2}{3}$	Start index of packed symmetric rank n .
$I(\alpha)$	Global packed index including ranks zero and one.
$J(\alpha) = I(\alpha) - 4$	Packed Taylor/derivative index, with ranks zero and one omitted.
M_α	Raw, unnormalized multipole moment.
$L_{ij,\alpha}$	Local Taylor derivative coefficient.
$D_{i_1 \dots i_n}$	$-\partial_{i_1} \dots \partial_{i_n} (1/r)$.
$A(\mu, \nu)$	$\mu(1 + \nu)/(1 - 2\nu)$.
E_v	Trace of the stored inclusion strain.
a	Sphere radius; in the direct routine, maximum of the three radii.
p, t	Multipole and Taylor truncation orders.